

1. Simplify the expression below into the form  $a + bi$ . Then, choose the intervals that  $a$  and  $b$  belong to.

$$\frac{-54.00 + 77.00i}{-8.00 - 3.00i}$$

- A.  $a \in [1.5, 3.5]$  and  $b \in [-11, -10.5]$   
B.  $a \in [6, 7]$  and  $b \in [-26.5, -25]$   
C.  $a \in [8.5, 10.5]$  and  $b \in [-6.5, -6]$   
D.  $a \in [1.5, 3.5]$  and  $b \in [-778.5, -777.5]$   
E.  $a \in [200, 201.5]$  and  $b \in [-11, -10.5]$
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2. Simplify the expression below into the form  $a + bi$ . Then, choose the intervals that  $a$  and  $b$  belong to.

$$(-8.00 - 5.00i)(4.00 - 3.00i)$$

- A.  $a \in [-53, -42]$  and  $b \in [1, 7]$   
B.  $a \in [-23, -14]$  and  $b \in [41, 47]$   
C.  $a \in [-23, -14]$  and  $b \in [-46, -43]$   
D.  $a \in [-53, -42]$  and  $b \in [-8, 3]$   
E.  $a \in [-33, -30]$  and  $b \in [9, 20]$
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3. Simplify the expression below and choose the interval the simplification is contained within.

$$7 - 6 \div 2 * 12 - (13 * 18)$$

- A.  $[-269, -262]$   
B.  $[-227.25, -222.25]$   
C.  $[234.75, 244.75]$   
D.  $[-762, -755]$

E. None of the above

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4. Choose the **smallest** set of Real numbers that the number below belongs to.

$$-\sqrt{\frac{1755}{15}}$$

- A. Whole
  - B. Integer
  - C. Rational
  - D. Irrational
  - E. Not a Real Number
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5. Choose the **smallest** set of Complex numbers that the number below belongs to.

$$\frac{\sqrt{210}}{11} + \sqrt{-8}i$$

- A. Rational
  - B. Irrational
  - C. Nonreal Complex
  - D. Pure Imaginary
  - E. Not a Complex Number
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6. Solve the equation below. Then, choose the interval that contains the solution.

$$-17(-16x - 14) = -6(11x - 19)$$

- A.  $x \in [-0.95, -0.15]$
- B.  $x \in [0.55, 1.5]$
- C.  $x \in [-1.2, -0.5]$

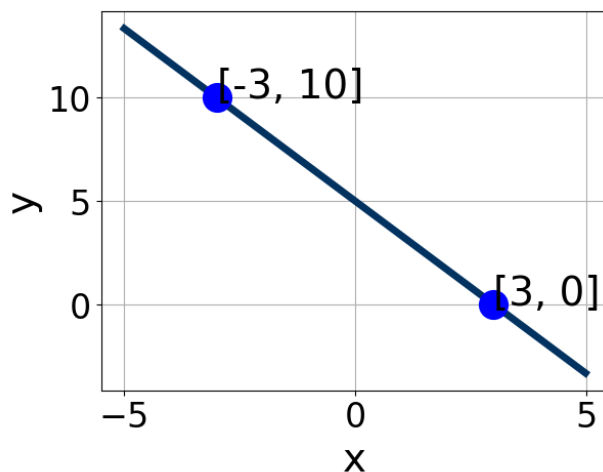
- D.  $x \in [-2.35, -1.6]$
- E. There are no Real solutions.

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7. Solve the linear equation below. Then, choose the interval that contains the solution.

$$\frac{7x + 8}{8} - \frac{8x - 4}{7} = \frac{3x - 9}{5}$$

- A.  $x \in [3.5, 4.5]$
- B.  $x \in [2, 3.5]$
- C.  $x \in [24, 25]$
- D.  $x \in [0.5, 1.5]$
- E. There are no Real solutions.

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8. Write the equation of the line in the graph below in Standard Form  $Ax + By = C$ . Then, choose the intervals that contain  $A$ ,  $B$ , and  $C$ .



- A.  $A \in [4.5, 5.5]$ ,  $B \in [2.5, 4]$ , and  $C \in [14.5, 16]$
- B.  $A \in [-6, -4]$ ,  $B \in [-3.5, -2.5]$ , and  $C \in [-15.5, -14]$
- C.  $A \in [4.5, 5.5]$ ,  $B \in [-3.5, -2.5]$ , and  $C \in [-15.5, -14]$
- D.  $A \in [0.5, 2]$ ,  $B \in [0.5, 2]$ , and  $C \in [3.5, 5.5]$
- E.  $A \in [0.5, 2]$ ,  $B \in [-2, 0.5]$ , and  $C \in [-5.5, -4.5]$

9. First, find the equation of the line containing the two points below. Then, write the equation in the form  $y = mx + b$  and choose the intervals that contain  $m$  and  $b$ .

$$(-8, 8) \text{ and } (-5, -6)$$

- A.  $a \in [-5.5, -4.5]$  and  $b \in [-30.5, -29]$
  - B.  $a \in [4, 6]$  and  $b \in [17, 18]$
  - C.  $a \in [-5.5, -4.5]$  and  $b \in [28.5, 30]$
  - D.  $a \in [-5.5, -4.5]$  and  $b \in [15.5, 16.5]$
  - E.  $a \in [-5.5, -4.5]$  and  $b \in [-2, -0.5]$
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10. Find the equation of the line described below. Write the linear equation in the form  $y = mx + b$  and choose the intervals that contain  $m$  and  $b$ .

Perpendicular to  $7x - 5y = 6$  and passing through the point  $(4, -3)$ .

- A.  $a \in [-1.3, -0.65]$  and  $b \in [-0.32, -0.05]$
  - B.  $a \in [0.2, 1.05]$  and  $b \in [-6.24, -5.62]$
  - C.  $a \in [-1.3, -0.65]$  and  $b \in [0.01, 0.21]$
  - D.  $a \in [-1.3, -0.65]$  and  $b \in [-7.2, -6.82]$
  - E.  $a \in [-1.5, -1.1]$  and  $b \in [-0.32, -0.05]$
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